



# **DIY Solar Power Systems Guide**

**Complete  
Handbook for  
Jamaica**

From phone-charging kits to  
10kW+ home arrays  
Learn to size, wire, and build  
solar systems

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# Welcome to Solar DIY

Solar energy is no longer just for large corporations or wealthy homeowners. With falling panel prices and improved battery technology, anyone can build their own solar power system.

## **Getting Started**

This guide covers systems from 100W portable kits to 10kW+ home arrays. All examples are tailored for Jamaica's climate with 5.0-5.5 peak sun hours daily.



## What You'll Learn

### Panels

Monocrystalline  
vs poly,  
wattage sizing,  
wiring configs

### Batteries

LiFePO4 vs  
lead-acid,  
capacity math,  
runtime

### Controllers

PWM vs MPPT,  
voltage  
matching,  
efficiency

### Inverters

Pure sine wave  
sizing, surge  
loads, AC  
distribution

# Small Systems

## (100W-500W)

Ideal for phone charging, laptop charging, LED TVs, and small electronics.

| System            | Use Case                                               | Cost Estimate |
|-------------------|--------------------------------------------------------|---------------|
| 100W Portable Kit | Charge 2 phones + 1 laptop daily; run LED TV 2 hours   | 50-50 USD     |
| 200W Starter      | Adds headroom for cloudy days. Great for sheds and RVs | 00-50 USD     |
| 400W Cabin Kit    | Small off-grid setup with inverter for                 | 00-00 USD     |

|  |                           |  |
|--|---------------------------|--|
|  | lights, fans,<br>charging |  |
|--|---------------------------|--|

**Pro Tip:** Start small. A 100W kit teaches you the basics without a big investment. You can always expand later.

# Medium Systems

## (500W-2kW)

Perfect for Starlink Mini, larger TVs, multiple laptops, and home office setups.

| System           | Use Case                                                                    |
|------------------|-----------------------------------------------------------------------------|
| 800W Office      | Power 2 laptops + TV + router during outages. Great for remote work.        |
| 1.2kW Hybrid     | Adds battery backup for evening use. Supports JPS net-billing if grid-tied. |
| 1.8kW Expandable | Room to add more panels later. Can handle small appliances plus devices.    |

## Key Components

**Panels:**

400W-1kW  
monocrystalline  
array

**Battery:** 12V

100Ah-200Ah  
LiFePO4 bank

**Controller:**

40A MPPT  
charge  
controller

**Inverter:**

1000W-  
2000W pure  
sine wave



# Large Systems (2kW-10kW+)

Full home power: multiple TVs, laptops, Starlink Mini, fridge, and all devices.

| System          | Use Case                                                                  |
|-----------------|---------------------------------------------------------------------------|
| 3kW Home Backup | Covers essential circuits: lights, wifi, fridge, TV, and laptop loads.    |
| 5kW Whole-Home  | Run most household circuits. Ideal for frequent outages or full off-grid. |
| 10kW+ Estate    | Multiple inverters, large battery bank, commercial-grade installation.    |

## Professional Installation:

Large systems require certified electricians. Nash Technologies provides full installation, JPS net-

billing support, and 15-year warranties.



## Safety First

**Electrical Safety Warning:** Solar systems produce high voltage DC that can cause serious injury or fire. Follow NEC code, use properly sized wire, install fuses/breakers on all conductors, and hire a licensed electrician for grid-tied connections.

## Wire Sizing & Fusing

Undersized wiring overheats. Size conductors for **1.25× expected current**. Fuse every positive conductor at the battery and between components. Use marine-grade or solar-rated cable for outdoor runs.

## NEC Requirements

**Disconnects:**

DC and AC disconnects required

**Grounding:**

All frames and enclosures grounded

**Labeling:**

Warning signs at all junction boxes

**Inspections:**

JPS inspection for grid-tied systems

# DIY Device Charging Systems

Step-by-step build guides for charging phones, laptops, Starlink Mini, and TVs with solar.

## Phone & Tablet Charging Kit

- **Panel:** 20W monocrystalline
- **Battery:** 12V 7Ah LiFePO4
- **Controller:** 10A PWM
- **Output:** 5V USB-A ports
- **Runtime:** 2-3 phones per day

## Laptop Charging Station

- **Panel:** 100W monocrystalline

- **Battery:** 12V 50Ah LiFePO4 (~640Wh)
- **Controller:** 20A MPPT
- **Inverter:** 300W pure sine wave
- **Runtime:** 2-4 hours laptop use

# More Device Setups

## Starlink Mini Power

- **Panel:** 120W monocrystalline
- **Battery:** 12V 100Ah LiFePO4
- **Controller:** 20A-30A MPPT
- **Inverter:** 300W pure sine wave
- **Runtime:** 6-10 hours off-grid

## TV & Entertainment

- **Panel:** 150W monocrystalline
- **Battery:** 12V 100Ah LiFePO4
- **Controller:** 20A MPPT
- **Inverter:** 500W pure sine wave
- **Runtime:** 3-6 hours for 55" LED

# Solar & Electrical

## Glossary

Simple definitions for common terms on solar labels, invoices, and JPS net-billing paperwork.

| Term                       | Definition                                                   |
|----------------------------|--------------------------------------------------------------|
| <b>Watt (W)</b>            | Unit of power. A 100W bulb uses 100 watts while on.          |
| <b>Kilowatt (kW)</b>       | 1,000 watts. A 5 kW system produces 5,000 watts at peak sun. |
| <b>Kilowatt-hour (kWh)</b> | Unit of energy. JPS bills in kWh.                            |
| <b>Volt (V)</b>            | Electrical pressure. Common solar banks: 12V, 24V, 48V.      |
| <b>Ampere (A)</b>          | Current flow. Wiring and fuses are sized by amps.            |



|                          |                                                                       |
|--------------------------|-----------------------------------------------------------------------|
| <b>Amp-hour (Ah)</b>     | Battery capacity. A 100Ah battery delivers 5A for 20 hours.           |
| <b>PSI</b>               | Pounds per square inch. Used for water pressure and pipe fittings.    |
| <b>BTU</b>               | British Thermal Unit. Heat energy; useful for water-heating sizing.   |
| <b>Peak Sun Hours</b>    | Hours of 1,000 W/m <sup>2</sup> irradiance. Jamaica averages 5.0–5.5. |
| <b>Inverter</b>          | Converts DC battery power to AC for appliances.                       |
| <b>Charge Controller</b> | Regulates panel-to-battery charging to prevent overcharge.            |
| <b>Grid-Tied</b>         | Connected to JPS lines. Allows net billing for excess energy.         |

# Solar Calculator

Estimate daily energy production and system cost for Jamaica conditions.

**Formula:**  $\text{Panel Watts} \times \text{Count} \times \text{Sun Hours} \times \text{Efficiency} \div 100$

| Input               | Value    |
|---------------------|----------|
| Panel Watts         | 400W     |
| Panel Count         | 4        |
| Sun Hours (Jamaica) | 5.5      |
| Efficiency          | 75%      |
| Daily Output        | 8,800 Wh |
| Estimated Cost      | ,280 USD |

Use this formula to size any system. Adjust panel count and wattage to match your energy needs.

# Solar Parts Finder

Select a system to see the full bill of materials with local-market pricing in JMD.

| Component  | Phone/<br>Tablet                       | Laptop<br>Station                        | Starlink<br>Mini                          |
|------------|----------------------------------------|------------------------------------------|-------------------------------------------|
| Panel      | 20W —<br>,000<br>JMD                   | 100W —<br>2,000<br>JMD                   | 120W —<br>4,000<br>JMD                    |
| Battery    | 12V<br>7Ah<br>LiFePO4<br>— ,500<br>JMD | 12V<br>50Ah<br>LiFePO4<br>— 8,000<br>JMD | 12V<br>100Ah<br>LiFePO4<br>— 5,000<br>JMD |
| Controller | 10A<br>PWM —<br>,500<br>JMD            | 20A<br>MPPT —<br>,000<br>JMD             | 20A–30A<br>MPPT —<br>4,000<br>JMD         |
| Inverter   | —                                      | 300W —<br>2,000<br>JMD                   | 300W —<br>2,000<br>JMD                    |

|              |                                                         |                                          |                                     |
|--------------|---------------------------------------------------------|------------------------------------------|-------------------------------------|
| Other        | 5V<br>USB-A<br>— ,200<br>JMD<br>Wiring<br>— ,500<br>JMD | Wiring<br>+<br>breakers<br>— ,000<br>JMD | DC-DC<br>converter<br>— ,000<br>JMD |
| <b>Total</b> | <b>~7,700<br/>JMD</b>                                   | <b>~3,000<br/>JMD</b>                    | <b>~0,000+<br/>JMD</b>              |

For full home systems and custom builds,  
contact Microtuff Solutions for a detailed  
quote.

## Need Help?

Professional installation, system design,  
and JPS net-billing support across  
Jamaica.

### **Nash Technologies Solutions**

Certified solar  
installation, 15-  
year  
warranties, free  
site  
assessments.

**1(876)  
359-9983**

### **Microtuff Solutions**

Security  
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803-6062**

**Ready to start your solar  
project?**

Contact us for a free consultation  
and site assessment. We serve all  
14 parishes across Jamaica.

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This guide is for educational purposes.  
Always consult a licensed electrician for  
installation.